R Lalith Vishnu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.05
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	93.80%
Gauthami Vidya Dhamam HS	Class X (Board of Secondary Education Telangana State)	2021	100.00%

WORK EXPERIENCE

Intern | ElectroChem Materials Group | IITH

- Engineered an automated hardware fabrication pipeline integrating **Solid Edge** models with **AxiDraw** using **Python** and **CMD** scripts.
- Reduced manual prototyping time by **~30%** while achieving **±0.005" precision** for interdigitated micro-electrode architectures.
- Co-authored a peer-reviewed publication in ***ACS Applied Electronic Materials*** by providing scripts and prototyping device fabrication.

Lab Safety Intern | IITH

- Retrieved many Safety manuals from top universities and contributed to drafting of IITH safety manual

Research Intern | Indian Institute of Science (IISc), Bangalore

- Engineered a **Python/PyVISA** control system, optimizing PID loops for **±0.003 °C** stability to capture high-fidelity **datasets**.
- Eliminated electrical noise via hardware shielding, enabling precise **picoampere** signal detection for clean **feature extraction**.
- Built a **5-20Hz SQLite** pipeline to automate thousands of testing cycles into structured, **ML-ready** time-series data.
- Developed a **GUI workflow** with real-time **Matplotlib** visualization and **Discord** alerts to ensure continuous **data ingestion**.

AI Systems Intern | Stremly

- Built an **MCP microservice** to interface with Nox (Stremly's core AI brain), enabling LLMs to execute agentic workflows via RESTful APIs.
- Integrated **Slack** and **Jira** via tool-calling, creating an "AI Product Manager" that helps in cross-app communication and team coordination.
- Secured HTTP infrastructure with cryptographic **signature verification** and Redis-based health monitoring.

PROJECTS

Lithium-Ion Battery Cycle Life Prediction Pipeline (GitHub Link):

- Parsed complex **HDF5**/MATLAB data using **h5py**, compressing 10GB+ raw files into a 3MB summary CSV.
- Mitigated sensor jitter via **rolling-mean** filtering and engineered features (**curvature, IR**), boosting **correlation** () from **-0.06 to -0.45**.
- Trained an **ElasticNet regression** model on early cycles (10-100), achieving **R^2=0.61** and ~275-cycle **RMSE** in lifespan prediction.

Automated C/C++ Windows Build Architecture for BMS Numerical Solvers:

- Engineered an OS-isolated Windows C/C++ environment using **CMake** and **MinGW**, enabling execution of complex

BMS simulations.
- Resolved MSVC/MinGW linker conflicts to seamlessly integrate high-performance **Intel MKL**, **OpenBLAS**, and SuperLU math solvers.
- Instrumented C code to profile solver latency, computing 360+ adaptive time steps and 500+ non-linear iterations in **~0.002 CPU seconds**.

Interactive AI Portfolio (Website Link)

- **Routed inference requests** across 8+ LLM APIs using Python asyncio; implemented a concurrent "Race" mode (FIRST_COMPLETED) to minimize Time-To-First-Token (**TTFT**) by returning the fastest stream and canceling pending network tasks.
- **Wrote a custom PyTorch inference engine** for Phi-3.5-Mini; implemented layer-by-layer weight loading and **KV caching** to execute local semantic RAG in under 500MB of RAM.
- **Streamed session telemetry** (LLM latency, IP geolocation, provider win-rates) to a Google Sheets sink via server-to-server OAuth (Service Accounts), fetching credentials dynamically at runtime from Azure Key Vault.

Urban Aerial Hierarchical Swarm Governance | Multi-Agent Systems Project (Git Hub Link) (ongoing)

- Architecting a **distributed optimization layer** using **ADMM** to enable decentralized, agentic decision-making across autonomous swarms.
- Formulating **multi-agent** constraints to autonomously negotiate complex traffic maneuvers, merging, and dynamic rerouting.
- Integrating decentralized agent workflows with hierarchical traffic controllers to achive zero-collision airspace via **ADMM convergence**.

PUBLICATIONS

Co-authored publication in **ACS** *Appl. Electron. Mater. on high-rate MXene-ionic liquid microsupercapacitors (DOI:* ***10.1021/acsaelm.6c01308****).*

HACKATHON

Mitsubishi 3D Tech Hackathon 2024 | Cash Prize Winner & Top 10 Finalist

- Engineered a **Docker-containerized LiDAR workflow**, with a synthetic data engine for 3D point cloud semantic segmentation.
- Automated **3D segmentation** using the **KPConv neural network with deformable kernels** pipeline on LiDAR point clouds.
- Built a **Blender Geometry Nodes** to process point clouds, achieving a **11× reduction in data preparation time**.

SIH Fin Shield Hackathon 2025 | National Finalists | Top 16 Teams

- Built an **end-to-end explainable AI platform** for credit risk prediction using **XGBoost** and **SHAP**.
- Developed a synthetic financial data pipeline using **Python, Pandas,** and **NumPy** to simulate real-world financial behavior.
- Trained and optimized the ML pipeline for key evaluation metrics including **AUC-ROC** and **F1-score**.
- Containerized and deployed the application using **Docker, Docker Compose,** and **REST APIs**.

POSITIONS OF RESPONSIBILITY & EXTRACURRICULAR ACTIVITIES

Coordinator, International Relations Cell

- Managed event PR outreach and co-designed official campus maps and delegation brochures.

Class Representative, Dept. of Industrial Chemistry - IIT Hyderabad

- Served as liaison for 22 students, resolving academic and administrative issues with faculty.

Core Team Member & Project Lead, Torque - IIT Hyderabad

- Promoted from team member to Core; Led R/C Car project and oversaw battery domain design for Mars Rover Challenge.

Operations Coordinator, Tinkerers' Lab - IIT Hyderabad

- Managed 80+ members, managing lab infrastructure, inventory, and procurement for sustainable operations.

Selected for Inter **IIT Tech Meet 2024**, worked on predicting EoL of Solar Panels and Industrial Solutions

Won the **NCC Best Shooter Award** among **400+ members** and completed **100+ hours of NSS community service**, including **Vidyadaan**.

CERTIFICATES AND LICENSEES

IBM | SkillBuild | From Learner to Builder: Become an AI Agent Architect

SKILLS

- **Languages & Core:** Python, SQL. *Familiar:* C/C++, JavaScript (Node.js/React.js).
- **AI, GenAI & Data:** XGBoost, CNN, SHAP, Scikit-learn, Llama 3.2, RAG Pipelines, LangChain, FAISS, GGUF (Quantization).
- **3D Vision & Hardware:** Open3D (Point Clouds, LiDAR), KPConv, SLAM, Blender (Geometry Nodes), Semantic Segmentation, PyVISA.
- **Systems & DevOps:** Docker, REST APIs, Model Context Protocol (MCP), Redis, SQLite, Streamlit, Git/GitHub.

Soft Skills: Adaptability, Public Speaking, Mentorship, Cross-team Collaboration, Analytical Thinking, Self-directed Learning, Networking